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Constraints on the Jurassic swing of the apparent polar wander path for the North China Block

Koji Uno[1] and Baochun Huang[2]

[1] *Department of Environmental Sciences, Faculty of Integrated Human Studies, Kyoto University, Japan. E-mail: unokoji@gaia.h.kyoto-u.ac.jp*
[2] *Institute of Geology and Geophysics, Chinese Academy of Sciences, Beijing, China*

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SUMMARY

In order to better constrain the Mesozoic apparent polar wander path for the North China Block, Late Triassic and Early Jurassic sandstones in the Ordos Basin and Early Jurassic sandstones in the Datong area were collected at 37 sites. Thermal demagnetization isolated a high-temperature magnetization component with unblocking temperatures of 500–580 °C from five sites of the Early Jurassic Yongdingzhuang Formation in the Datong area. Palaeomagnetic directions from the five sites yielded a positive fold test at the 95 per cent confidence level. These directions combined together with previously reported ones give a new Early Jurassic mean direction ($D = 358.0^\circ$, $I = 50.6^\circ$, $\alpha_{95} = 4.4^\circ$) and corresponding palaeomagnetic pole (81.5°N, 302.5°E, $A_{95} = 5.5^\circ$). The palaeomagnetic pole is regarded as representative of the North China Block because this pole is derived from two localities in the undisputed part of the North China Block. Comparisons of the Early Jurassic pole for the North China Block with coeval ones for the South China Block identify significant differences between the poles from the two regions. At Early Jurassic times, the North China Block was at the peak of rotational motion associated with the collision of the North and South China Blocks. The North China Block underwent a counterclockwise rotation of 12°–32° with respect to the South China Block since the Early Jurassic, at least 17 per cent of the total rotational motion remaining at that time.

Key words: Mesozoic, North China Block, palaeomagnetism, tectonics.

1 INTRODUCTION

The North China Block and the South China Block are the major tectonic blocks forming East Asia (Fig. 1). The two blocks were located in distinct positions from each other through Precambrian–Palaeozoic times and were welded through a collision process in Early Mesozoic times. The collision of the North and South China Blocks is recognized by metamorphic belts observed between the two blocks. The Dabie Belt and the Sulu Belt that bound the two blocks yield coesite-bearing ultrahigh-pressure metamorphic rocks (Okay & Şengör 1992). The Tan-Lu Fault also marks the boundary between the North and South China Blocks. This fault, to the north of the Dabie Belt and to the south of the Sulu Belt, marks the biogeographical and lithological boundary between the two blocks (Kobayashi 1967).

Permian and Early Mesozoic palaeomagnetic data sets from the North and South China Blocks suggest that the two blocks sutured following a scissor-like collision process (Zhao & Coe 1987). The North and South China Blocks experienced counterclockwise and clockwise rotations during the welding event, respectively. The North China Block rotated counterclockwise by more than 60° relative to the South China Block since Late Permian times about a pole of rotation located in the eastern part of the two blocks. The scissor-like process implied westward progression of the collision. Geological observations supported the westward closure of an ocean between the North and South China Blocks (Yin & Nie 1993).

The Mesozoic apparent polar wander path of the North China Block is characterized by two tracks (Gilder & Courtillot 1997). One is Early Triassic to Middle–Late Jurassic in age, and another is from Middle–Late Jurassic to Cretaceous. The former track is the largest segment in the Mesozoic apparent polar wander path for the North China Block: the palaeomagnetic poles for this block in the Early Triassic lie around 50–60°N, 350°E (Yang *et al.* 1991; Embleton *et al.* 1996), while those in the Middle and Late Jurassic fall around 70–80°N, 220–230°E (Yang & Besse 2001). The pole position shifted by more than 40° in this period. Gilder & Courtillot (1997) interpreted that the termination of this earlier segment means cessation of the collision event and that the bulk of tectonic motion of the North China Block due to the collision occurred between the Early Triassic and Middle–Late Jurassic.

Late Triassic and Early Jurassic palaeomagnetic poles from the North China Block are critical with respect to the determination of the largest segment in the Mesozoic apparent polar wander path.

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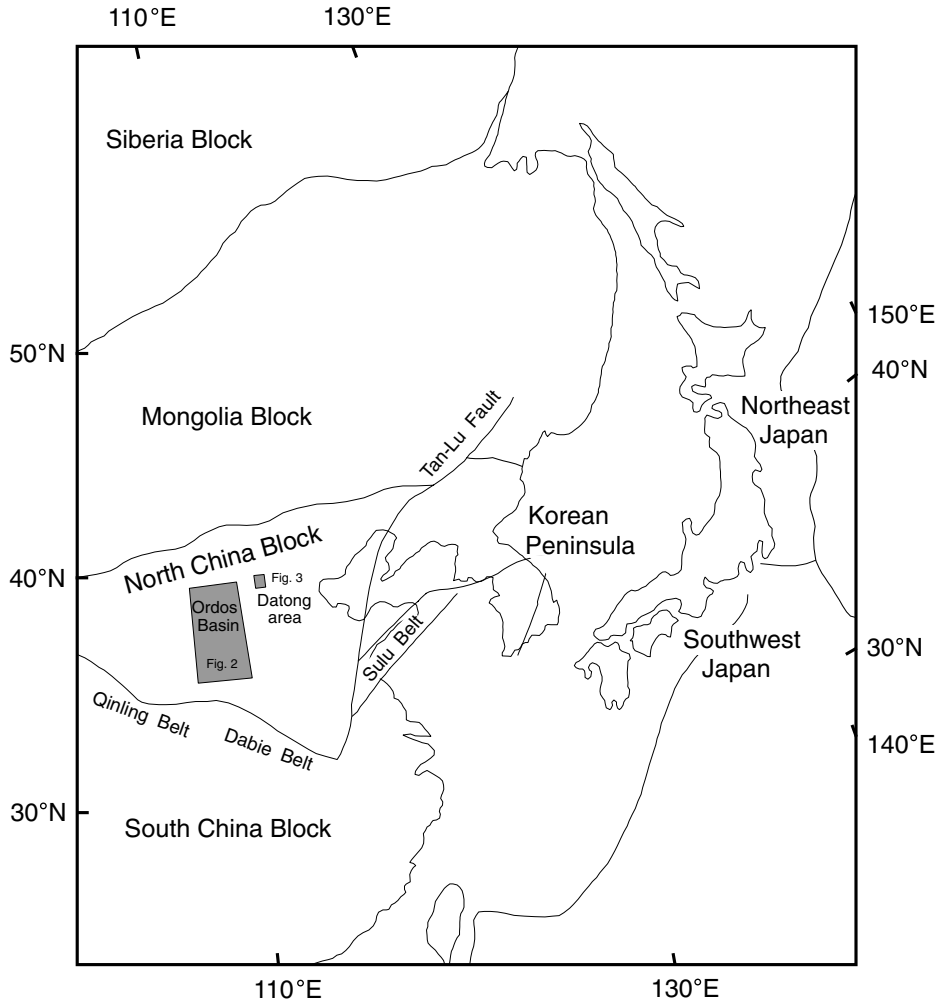


Figure 1. (a) Outline tectonic map of East Asia showing the study areas (Ordos Basin and Datong area) in the North China Block.

These poles record the precise relative rotation of the North China Block with respect to the South China Block during the welding event. At present, the Late Triassic and Early Jurassic representative poles for the North China Block are based on palaeomagnetic data from a small number of studies (Yang *et al.* 1991, 1992) that come from a single locality within the block (Ordos Basin, Fig. 1). The palaeomagnetic data of these important time periods have not been improved over the last decade. Attempts to accumulate reliable Triassic and Jurassic data have been thwarted by remagnetizations and lack of outcrops with good structural conditions to perform fold tests. As seen in the Early Jurassic demagnetization results of Yang *et al.* (1992), even successful palaeomagnetic works have encountered difficulties in isolating a high-fidelity record of geomagnetic field directions in the North China Block, and a further study needs to be undertaken in order to contribute toward establishing representative Triassic and Jurassic poles for the North China Block.

In this paper we present new palaeomagnetic results from Late Triassic and Early Jurassic rocks in the North China Block. This paper includes the first report of a positive fold test for Early Jurassic rocks in the North China Block and improves the representative pole position in the Early Jurassic.

2 GEOLOGICAL SETTING AND SAMPLING

The undisputed part of the North China Block is generally regarded as an area south of the Mongolia Block, north of the Qinling–Dabie Belt and west of the Tan-Lu Fault (Fig. 1). This block, together with the Korean Peninsula, has been also referred to as the Sino-Korean Block. Archaean rock units predominantly compose the basement of the North China Block (Jahn *et al.* 1988; Song *et al.* 1996). The basement is unconformably overlain by well-developed Cambrian to Middle Ordovician, Upper Carboniferous to Permian and Mesozoic strata.

The Ordos Basin is a large continental sedimentary basin located in the western part of the North China Block, where Late Palaeozoic to Cretaceous sediments are distributed. Triassic and Jurassic strata are well exposed in the eastern part of the Ordos Basin (Fig. 2). The Upper Triassic is divided into three formations, the Hujiacun, Yongping and Wayaobao Formations in ascending order. They consist of greyish to greenish fine sandstones and sandy shales with interbedded coal seams, and contain plant fossils of *Danaeopsis fecunda* and *Neocalamites carcinoides* (Bureau of Geology and Mineral Resources of Shaanxi Province 1989). The Early Jurassic Fuxian

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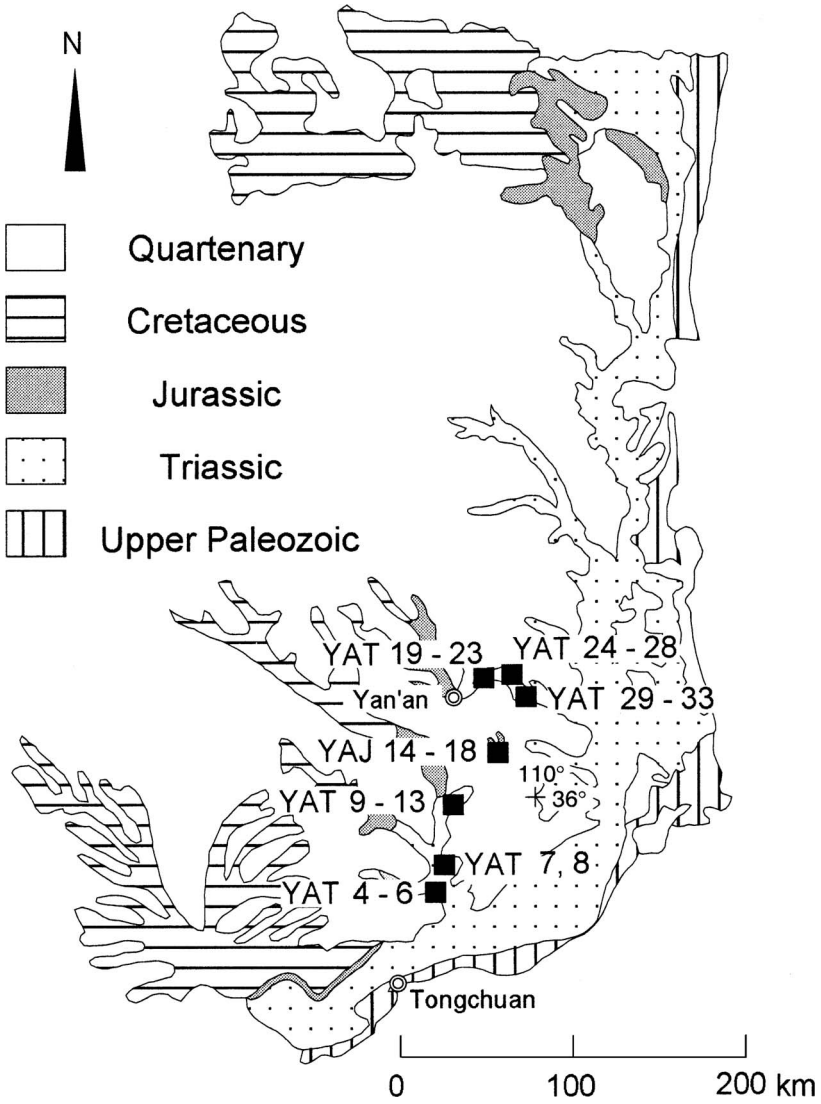


Figure 2. Geological map of the Ordos Basin (Bureau of Geology and Mineral Resources of Shaanxi Province 1989). Solid squares denote the palaeomagnetic sampling sites.

Formation crops out to the west of the Late Triassic formations. It is made up mainly of purplish to greenish fine sandstones and sandy shales, containing elements of *Eosolimnadiopsis gushanensis* and *Ferganoconcha subcentralis*.

The Datong area is located to the northeast of the Ordos Basin, where Late Palaeozoic and Jurassic to Cretaceous sediments are distributed (Fig. 3). In this area, the Early Jurassic Yongdingzhuang Formation composes southern peripheries of the Jurassic sedimentary basin. It is composed of greyish to purplish sandstones and sandy shales, and contains plant fossils of *Coniopteris hymenophylloides, Czekanowskia rigida* and *Cladophlebis shansiensis* (Bureau of Geology and Mineral Resources of Shanxi Province 1989).

In the Ordos Basin, we collected pale to dark grey sandstones in the Late Triassic Hujiacun, Yongping and Wayaobao Formations at 25 sites and pale grey sandstones in the Early Jurassic Fuxian Formation at five sites (Fig. 2). In the Datong area, grey sandstones in the Yongdingzhuang Formation were collected at seven sites (Fig. 3). The bedding attitudes in the Ordos Basin are flat (11 sites) or gentle dips ranging from 3° to 10° (19 sites), while those in the Datong area show northward dips of approximately 20°–30° (Table 1). Folding in the Datong area is inferred to have occurred during the Middle Jurassic on the basis of an obvious structural discordance between the Middle Jurassic Datong and Yungang Formations (Zheng *et al.* 1991). For each site, between 10 and 13 cores were collected using a portable drill and oriented using a magnetic compass. The Late Triassic sampling sites in the present study cover a distance of approximately 130 km in the Ordos Basin and are distributed to the north of the previous sampling sites (35.2°N, 109.1°E, Yang *et al.* 1991). Our sampling sites of the Early Jurassic sediments in the Ordos Basin are located approximately 40 km away from the previous ones (36.1°N, 109.3°E, Yang *et al.* 1992). Palaeomagnetic sampling of Early Jurassic sediments was previously performed at one site in the Datong area; however, only unsuccessful palaeomagnetic results were obtained (Zheng *et al.* 1991).

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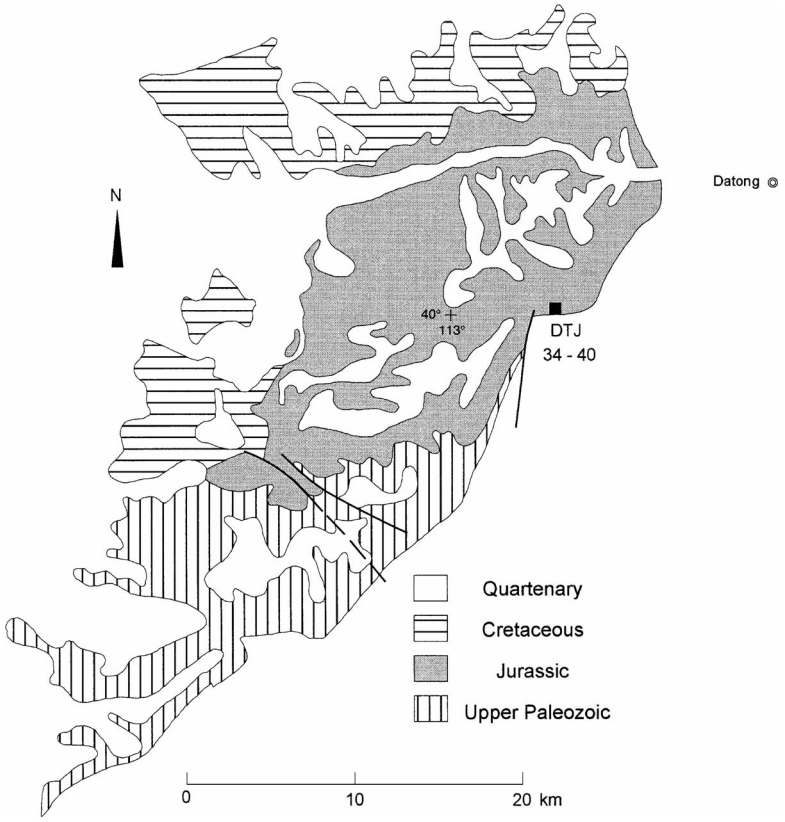


Figure 3. Geological map of the Datong area (Bureau of Geology and Mineral Resources of Shanxi Province 1989). Solid square denotes the palaeomagnetic sampling sites.

3 PALAEOMAGNETISM

Natural remanent magnetizations (NRMs) were measured with a 2-G superconducting rock magnetometer (SRM) at Kyoto University. Samples were subjected to progressive thermal demagnetization treatment with a Natsuhara TDS-1 thermal demagnetizer. Results for each sample were plotted on orthogonal vector diagrams (Zijderveld 1967) to evaluate the demagnetization behaviour. Principal component analysis (Kirschvink 1980) techniques were used to estimate directions of the magnetic components (Table 2). Thermal demagnetization of a three-component isothermal remanent magnetization (IRM) (Lowrie 1990) was performed with a Magnetic Measurements pulse magnetizer to identify the ferromagnetic mineral content.

Two magnetization components are recognized at five sites (DTJ34, 35, 36, 37 and 38, see Figs 4a–d), while a single magnetization component is observed in 28 sites (Figs 4e and f). The remaining four sites (YAT6, 19, 21 and DTJ39) exhibit only erratic directional behaviour with no stable endpoints during demagnetization. All of the five sites with two-component magnetization are of Early Jurassic age and located in the Datong area. The sites with a single-component magnetization, of Late Triassic and Early Jurassic ages, are located in the Ordos Basin except one Early Jurassic site, which is in the Datong area (DTJ40).

3.1 Samples with two remanent magnetization components

The low-temperature magnetization component is identified at the initial stages of thermal demagnetization. A large fraction of this component is demagnetized by 200 °C. The low-temperature component generally has unblocking temperatures of between 250 and 350 °C. Following removal of the low-temperature component, the high-temperature component is isolated. The high-temperature component is unblocked at approximately 500–580 °C. The maximum observed unblocking temperature of 580 °C suggests that the high-temperature component resides in magnetite. Thermal demagnetization of the three-component IRM was performed on representative samples, showing an unblocking temperature of 580 °C. An unblocking temperature of 580 °C is revealed (Fig. 5) and indicates that the magnetic carrier is magnetite.

The site-mean directions of the low-temperature component have precise parameters (k-value) ranging from 107.0 to 253.8 (Table 2). The mean directions of the five sites before and after tilt correction are $D_g = 355.4^\circ$, $I_g = 57.2^\circ (\alpha_{95} = 1.7^\circ)$ and $D_s = 351.2^\circ$, $I_s = 29.5^\circ (\alpha_{95} = 3.9^\circ)$, respectively (Fig. 6a). The former mean direction is parallel to the present Earth's field direction ($D = 355.1^\circ$, $I = 58.9^\circ$). These directions yield a negative fold test (McFadden 1990) at the 95 per cent confidence level (*in situ* $\xi_2 = 2.16$, tilt-corrected $\xi_2 = 4.01$, $\xi = 2.61$, critical value at 95 per cent confidence). We interpret the low-temperature component as being acquired in recent times.

The directions of the high-temperature component cluster well at each site. The precision parameter of the five site-mean directions ranges between 17.8 and 130.6. Before tilt correction these directions, with a mean of $D_g = 7.2^\circ$, $I_g = 75.5^\circ$ with $\alpha_{95} = 5.2^\circ$, have a relatively steep inclination (Fig. 6b). We observe a clear difference in direction between the high- and low-temperature components. The site-mean directions of the high-temperature component are

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Table 1. Palaeomagnetic sampling details of Late Triassic and Early Jurassic rocks of the North China Block.

Site	Area	Age	*N*	Formation	Locality		Strike dip	
					Lat. (°N)	Long. (°E)	(deg)	(deg)
YAT4	Ordos Basin	Late Triassic	10	Hujiacun	35.598	109.323	0	0
YAT5	Ordos Basin	Late Triassic	11	Hujiacun	35.597	109.322	0	0
YAT6	Ordos Basin	Late Triassic	11	Hujiacun	35.599	109.323	0	0
YAT7	Ordos Basin	Late Triassic	11	Hujiacun	35.605	109.351	0	0
YAT8	Ordos Basin	Late Triassic	11	Hujiacun	35.605	109.351	0	0
YAT9	Ordos Basin	Late Triassic	10	Wayaobao	36.030	109.451	230	4
YAT10	Ordos Basin	Late Triassic	11	Wayaobao	36.030	109.451	230	4
YAT11	Ordos Basin	Late Triassic	11	Wayaobao	36.030	109.451	191	4
YAT12	Ordos Basin	Late Triassic	10	Wayaobao	36.029	109.450	0	0
YAT13	Ordos Basin	Late Triassic	10	Wayaobao	36.029	109.450	0	0
YAJ14	Ordos Basin	Early Jurassic	13	Fuxian	36.343	109.706	322	8
YAJ15	Ordos Basin	Early Jurassic	12	Fuxian	36.344	109.706	322	8
YAJ16	Ordos Basin	Early Jurassic	10	Fuxian	36.343	109.706	126	6
YAJ17	Ordos Basin	Early Jurassic	10	Fuxian	36.343	109.707	265	3
YAJ18	Ordos Basin	Early Jurassic	10	Fuxian	36.343	109.707	265	3
YAT19	Ordos Basin	Late Triassic	10	Wayaobao	36.696	109.704	258	6
YAT20	Ordos Basin	Late Triassic	10	Wayaobao	36.696	109.704	218	4
YAT21	Ordos Basin	Late Triassic	10	Wayaobao	36.696	109.704	241	3
YAT22	Ordos Basin	Late Triassic	10	Wayaobao	36.696	109.705	174	4
YAT23	Ordos Basin	Late Triassic	10	Wayaobao	36.696	109.705	228	10
YAT24	Ordos Basin	Late Triassic	10	Yongping	36.710	109.836	245	6
YAT25	Ordos Basin	Late Triassic	10	Yongping	36.710	109.836	245	6
YAT26	Ordos Basin	Late Triassic	10	Yongping	36.710	109.835	267	6
YAT27	Ordos Basin	Late Triassic	10	Yongping	36.710	109.835	267	6
YAT28	Ordos Basin	Late Triassic	10	Yongping	36.711	109.835	0	0
YAT29	Ordos Basin	Late Triassic	10	Hujiacun	36.599	109.907	0	0
YAT30	Ordos Basin	Late Triassic	10	Hujiacun	36.599	109.907	235	3
YAT31	Ordos Basin	Late Triassic	11	Hujiacun	36.599	109.907	324	4
YAT32	Ordos Basin	Late Triassic	10	Hujiacun	36.599	109.906	0	0
YAT33	Ordos Basin	Late Triassic	10	Hujiacun	36.599	109.906	0	0
DTJ34	Datong area	Early Jurassic	10	Yongdingzhuang	40.004	113.117	257	30
DTJ35	Datong area	Early Jurassic	11	Yongdingzhuang	40.004	113.117	257	30
DTJ36	Datong area	Early Jurassic	10	Yongdingzhuang	40.004	113.117	257	30
DTJ37	Datong area	Early Jurassic	10	Yongdingzhuang	40.004	113.117	252	20
DTJ38	Datong area	Early Jurassic	10	Yongdingzhuang	40.004	113.117	247	30
DTJ39	Datong area	Early Jurassic	11	Yongdingzhuang	40.005	113.117	243	33
DTJ40	Datong area	Early Jurassic	11	Yongdingzhuang	40.004	113.117	243	33

N, The number of the samples collected; Lat., north latitude; Long., east longitude; strike, right-hand strike.

significantly separate from those of the low-temperature component before tilt correction.

The directions of the high-temperature component, after tilt correction, are of northerly declination and intermediate positive inclination (Fig. 6b). The mean direction of the five sites after tilt correction is $D_s = 352.7^\circ$, $I_s = 48.3^\circ$($\alpha_{95} = 3.2^\circ$). Tilt correction increases the precision parameter of the mean direction from to 219.3 to 577.8 (Table 2). A positive fold test (McFadden 1990) is observed at the 95 per cent confidence level (*in situ* $\xi_2 = 3.65$, tilt-corrected $\xi_2 = 1.69$, $\xi = 2.61$, critical value at 95 per cent confidence), indicating that the high-temperature component is of pre-folding origin. The high-temperature component is interpreted to be of primary origin because the folding age is Middle Jurassic. The primary nature of this direction is further confirmed by the obvious disagreement with the Middle Jurassic direction from the same area ($D = 17.3^\circ$, $I = 56.7^\circ$ with $\alpha_{95} = 2.5^\circ$, Zheng *et al.* 1991).

3.2 Samples with a single remanent magnetization component

These samples are dominated by a single magnetization component at initial stages of thermal demagnetization. The unblocking temperatures of this component concentrate between 200 and 300 °C (Figs 4e and f). Above these temperatures, the demagnetization behaviour becomes erratic and higher unblocking temperature components are not recoverable. The directions of the low-temperature component before tilt correction, with a mean of $D_g = 355.5^\circ$, $I_g = 52.8^\circ$ with $\alpha_{95} = 3.4^\circ$, conform to the present Earth's field direction ($D = 355.1^\circ$, $I = 58.9^\circ$) (Fig. 6c). This component is considered to be a present-day viscous overprint.

4 DISCUSSION

4.1 Early Jurassic palaeomagnetic directions from the North China Block

The Early Jurassic primary magnetization component with a positive fold test was first obtained from the Yongdingzhuang Formation in the Datong area. The previous palaeomagnetic studies of Early Jurassic rocks from the North China Block were performed by Zheng *et al.* (1991) and Yang *et al.* (1992). Zheng *et al.* (1991) attempted to isolate a primary component from the

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Table 2. Site mean palaeomagnetic results from the Early Jurassic rocks of the Datong area.

Site	N	*In situ*		Tilt corrected			
		D (deg)	I (deg)	D (deg)	I (deg)	k	α_{95} (deg)
High-temperature component							
DTJ34	4	341.1	77.6	345.2	47.7	47.5	13.5
DTJ35	7	14.0	78.0	355.4	48.7	29.9	11.2
DTJ36	9	13.2	76.2	356.0	47.2	17.8	12.6
DTJ37	8	8.2	67.6	356.8	48.8	130.6	4.9
DTJ38	6	16.4	76.3	350.2	48.6	33.3	11.8
Mean	5	7.2	75.5			219.3	5.2
				352.7	48.3	577.8	3.2
J1 Mean	15			358.0	50.6	76.3	4.4
Low-temperature component							
DTJ34	7	355.9	58.5	352.3	28.8	253.8	3.8
DTJ35	9	355.9	56.8	352.5	27.0	137.3	4.4
DTJ36	9	352.1	58.3	350.1	28.3	154.1	4.2
DTJ37	7	354.4	56.3	350.4	36.6	157.6	4.8
DTJ38	8	358.7	55.8	350.6	27.0	107.0	5.4
Mean	5	355.4	57.2			2071.0	1.7
				351.2	29.5	382.6	3.9

N, the number of the samples in calculation for the means; D and I, declination and inclination, respectively; k, the precision parameter; α_{95}, radius of the cone of the 95 per cent confidence.
The Early Jurassic (J1) mean direction is calculated by combining the present data with those of Yang *et al.* (1992), at the Datong area reference point (40.0°N, 113.0°E).

Yongdingzhuang Formation, but they failed to discern any magnetic component due to unstable behaviour of remanences during thermal demagnetization. Yang *et al.* (1992) presented Early Jurassic directions from the Fuxian Formation at ten sites in the Ordos Basin. These directions possessed both normal and reversed polarities that are statistically antipodal; however, they have large confidence limits ranging from 9.8° to 29.3°, with 60 per cent of the sites showing α_{95} larger than 20° (Fig. 6d). The large uncertainty in direction needs to be improved by a coeval data set with a positive fold test, because Early Jurassic palaeomagnetic directions for the North China Block are expected to be close to the present Earth's field direction, and the Early Jurassic directions by Yang *et al.* (1992) reveal a slight dispersion after tilt correction (*in situ* $k = 60.4$, tilt-corrected $k = 52.1$).

By combining the present data with those of Yang *et al.* (1992), an Early Jurassic palaeomagnetic direction that is representative of the North China Block can be calculated. Since the previous data only come from the Ordos Basin, there is uncertainty as to whether or not results from the Ordos Basin are representative of the entire North China Block. A good agreement in direction is observed between data sets of the two studies (Fig. 6d). It is therefore concluded that a new data set derived from two localities (Datong area and Ordos Basin) is regarded as being representative of this block. The palaeomagnetic directions of the present study and those of Yang *et al.* (1992) are averaged to yield a mean direction of $D_s = 358.0°$, $I_s = 50.6°$($\alpha_{95} = 4.4°$) at the reference location of the Datong area (40.0°N, 113.0°E) and a corresponding palaeomagnetic pole at 81.5°N, 302.5°E ($A_{95} = 5.5°$). The mean direction covers a distance of approximately 530 km within the interior part of the North China Block defined by Gilder & Courtillot (1997), an area south of the Mongolia Block, north of the Qinling–Dabie Belt and west of the Tan-Lu Fault (Fig. 1).

Table 3. Palaeomagnetic poles for the North and South China Blocks.

Age	N (n)	Pole			Reference
		Lat. (°N)	Long. (°E)	A_{95} (deg)	
North China Block					
J2	6	76.4	234.8	5.1	Yang & Besse (2001)
J1	(15)	81.5	302.5	5.5	This study
Tr3	(11)	62.3	7.7	3.8	Yang *et al.* (1991)
Tr2	7	61.0	2.0	2.9	Yang & Besse (2001)
South China Block					
J1-2	5	79.9	221.8	5.3	Yang & Besse (2001)
J1-2	3	64.3	193.5	9.0	Gilder & Courtillot (1997)
Tr2	8	42.7	212.6	6.0	Yang & Besse (2001)

Tr2, Middle Triassic; Tr3, Late Triassic; J1, Early Jurassic; J1–2, Early to Middle Jurassic; J2, Middle Jurassic;
N (n), the number of the studies (sites); Lat., north latitude; Long., east longitude; A_{95}, radius of the circle of the 95 per cent confidence.

4.2 Jurassic apparent polar wander path for the North China Block

There appear to be two groups of Mesozoic pole positions for the North China Block. One is of Early to Late Triassic ages located around 60°N latitude, 0°E longitude, another is of Middle Jurassic to Cretaceous age located around 70–80°N latitude, 200–230°E longitude (Zhao *et al.* 1996; Gilder & Courtillot 1997; Yang & Besse 2001). The largest track in the Mesozoic apparent polar wander path is observed as the arc that connects the two groups of poles, while another track is relatively small and exists within the younger group of poles (Gilder & Courtillot 1997). During Mesozoic times the North China Block shifted its pole position by approximately 40°.

The Early Jurassic pole of the present study is situated alone on the large track (Table 3, Fig. 7). The Early Jurassic pole is significantly removed by 25.2° ± 6.7° from the Late Triassic pole for the same block (62.3°N, 7.7°E with $A_{95} = 3.8°$, Yang *et al.* 1991) and by 13.0° ± 7.5° from the Middle Jurassic one (76.4°N, 234.8°E with $A_{95} = 5.1°$, Yang & Besse 2001). This Early Jurassic pole constrains the Mesozoic apparent polar wander path for the North China Block, because it is the only pole that does not belong to the two major populations of Mesozoic poles. At Early Jurassic time, the North China Block was at the peak of rotational motion associated with the collision of the North and South China Blocks.

On the other hand, two distinct Early–Middle Jurassic pole positions have been proposed for the South China Block (Fig. 7). Gilder & Courtillot (1997) presented an Early–Middle Jurassic pole (64.3°N, 193.5°E with $A_{95} = 9.0°$) on the basis of poles from three studies. Yang & Besse (2001) calculated a mean pole of this period from five studies (79.9°N, 221.8°E with $A_{95} = 5.3°$). A major factor relevant to the angular distance between the two poles (17.4° ± 10.4°) is the correction of local rotation of poles. An Early Jurassic pole reported by Gilder *et al.* (1993), for example, is included in the data set of Gilder & Courtillot (1997) after correction of local tectonic rotation, whereas Yang & Besse (2001) added the pole into their data set without correction. Although each study aimed at diminishing the effects of local tectonic rotation, the pole position in this period remains debatable. We prefer to regard both poles as being representative of the South China Block during the Early–Middle Jurassic.

The Early Jurassic pole for the North China Block differs by 29.5° ± 10.5° from the coeval South China Block pole of Gilder & Courtillot (1997) and by 12.1° ± 7.6° from that of Yang & Besse (2001). The significant separation of poles between the North and

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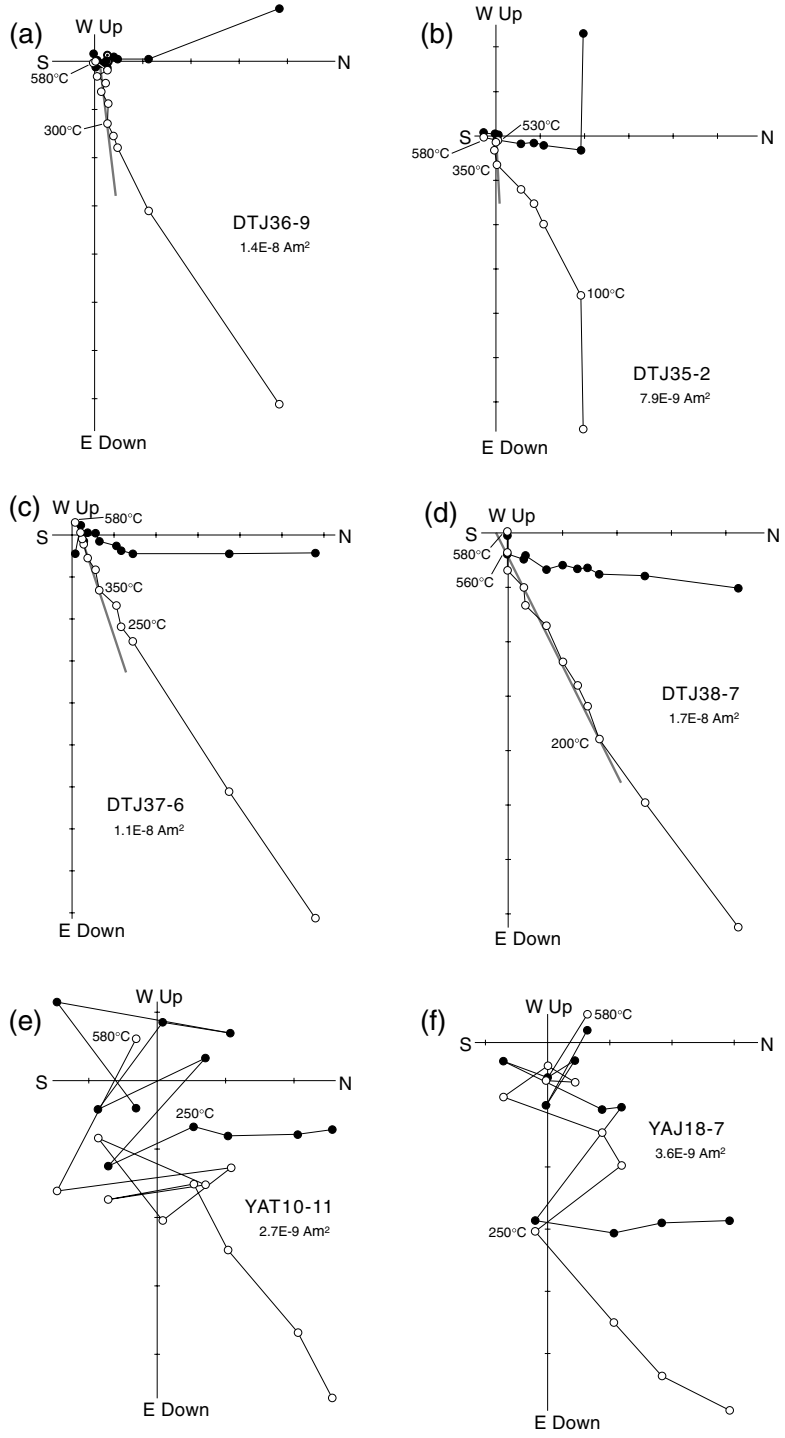


Figure 4. Orthogonal plots of thermal demagnetization results (before tilt correction) of the Early Jurassic rocks in the Datong area (a–d), Late Triassic rocks in the Ordos Basin (e) and Early Jurassic rocks in the Ordos Basin (f). Solid and open symbols in demagnetization diagrams represent projections on horizontal and vertical planes, respectively.

South China Blocks corroborates that the relative tectonic movement between the two blocks was not completed by the Early Jurassic. Yang & Besse (2001) recently proposed a possibility that the welding of the two blocks ceased during the Early Jurassic, based on an intersection of error limits between the Early Jurassic pole from the North China Block and the Early–Middle Jurassic one from the South China Block. However, the Early Jurassic North China Block pole of the present study is determined with precision and substantiates that relative movement between the two blocks occurred after the Early Jurassic.

The difference in pole position between the North and South China Blocks is ascribed to counterclockwise rotation of the North

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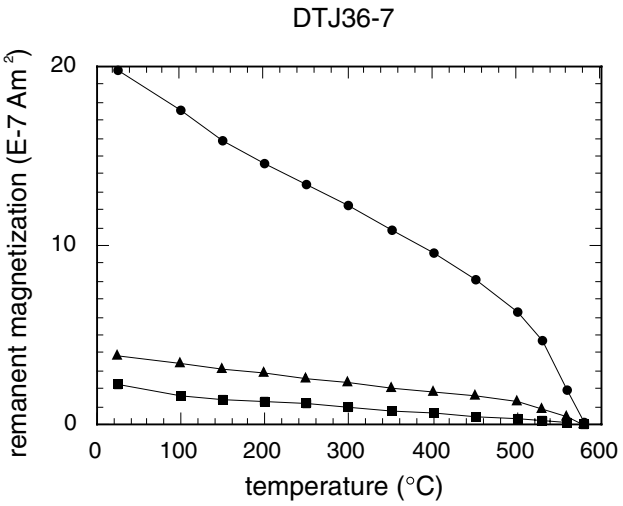


Figure 5. Thermal demagnetization of a three-component IRM (Lowrie 1990): circles indicate the soft component, 0.12 T field; triangles indicate the medium component, 0.4 T field; squares indicate the hard component, 2.6 T field.

China Block with respect to the South China Block, because no significant palaeolatitude contrast is present between the two blocks at the Datong area reference point (40.0°N, 113.0°E). The pole for the North China Block is rotated 14°–35° counterclockwise with respect to those of the South China Block at the reference point. Since a pivot point of rotation of the North and South China Blocks has been previously proposed, values of rotation at the point should be evaluated. According to Gilder & Courtillot (1997), the pole of rotation of the two blocks lies at around 31.6°N, 116.0°E. The North China Block shows counterclockwise rotation of 12°–32° with respect to the South China Block calculated at the pole of rotation, and no significant palaeolatitude difference exists between the two regions. Between 17 and 44 per cent of the relative rotation is considered to have occurred after the Early Jurassic, because the amount of relative rotation between the North and South China Blocks since the Early Triassic is estimated to be 71° (Zhao *et al.* 1996).

Evaluation of Early Jurassic palaeomagnetic data shows that at least 17 per cent of the total rotation was left at that time. The collision process eventually ended during the Middle–Late Jurassic (Gilder & Courtillot 1997) or the Early Cretaceous (Yokoyama *et al.* 2001). The independent age estimates from isotopic data, however, suggested earlier termination of the collision between the two blocks. Isotopic data of high- to ultrahigh-pressure metamorphism (Ames *et al.* 1993, 1996; Li *et al.* 1993; Rowley *et al.* 1997; Hacker *et al.* 1998) and exhumation-related cooling (Eide *et al.* 1994; Eide & Liou 2000) in the Qinling–Dabie–Sulu Belt have ages of Triassic (245–210 Ma) and Late Triassic–Early Jurassic (220–190 Ma), respectively, predating the palaeomagnetic age estimates. The bulk of metamorphic and cooling ages from the Dabie Belt are between 225 and 190 Ma (Late Triassic to Early Jurassic times) (Eide & Liou 2000). Because the Dabie Belt is considered to have been located near the pivot point of rotation, this belt has probably experienced the entire collision process, including the whole relative rotation process. The peak in isotopic ages is therefore interpreted as the time of rapid relative rotation (the peak in angular velocity) between the North and South China Blocks (Gilder & Courtillot 1997). The tectonic feature of the North China Block during Early Jurassic times, and Late Triassic times, is characterized by large rotational motion with a counterclockwise sense relative to the South China Block.

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REFERENCES

Ames, L., Tilton, G.R. & Zhou, G., 1993. Timing of collision of the Sino-Korean and Yangtze cratons: U-Pb zircon dating of coesite-bearing eclogites, *Geology,* **21,** 339–342.

Ames, L., Zhou, G. & Xiong, B., 1996. Geochronology and isotopic character of ultrahigh-pressure metamorphism with implications for collision of the Sino-Korean and Yangtze cratons, central China, *Tectonics,* **15,** 472–489.

Bureau of Geology and Mineral Resources of Shaanxi Province, 1989. *Regional Geology of Shaanxi Province,* Geol. Publ. House, Beijing.

Eide, E.A. & Liou, J.G., 2000. High-pressure blueschists and eclogites in Hong'an: a framework for addressing the evolution of high- and ultrahigh-pressure rocks in central China, *Lithos,* **52,** 1–22.

Eide, E.A., McWilliams, M.O. & Liou, J.G., 1994. $^{40}Ar/^{39}Ar$ geochronology and exhumation of high-pressure to ultrahigh-pressure metamorphic rocks in east central China, *Geology,* **22,** 601–604.

Embleton, B.J.J., McElhinny, M.W., Ma, X., Zhang, Z. & Li, Z.X., 1996. Permo-Triassic magnetostratigraphy in China: the type section near Taiyuan, Shanxi Province, North China, *Geophys. J. Int.,* **126,** 382–388.

Gilder, S. & Courtillot, V., 1997. Timing of the North-South China collision from new middle to Late Mesozoic paleomagnetic data from the North China Block, *J. geophys. Res.,* **102,** 17 713–17 727.

Gilder, S.A., Zhao, X., Coe, R.S., Wu, H. & Kuang, G., 1993. Discordance of Jurassic paleomagnetic data from south China and their tectonic implications, *Earth planet. Sci. Lett.,* **119,** 259–269.

Hacker, B.R., Ratschbacher, L., Webb, L., Ireland, T., Walker, D. & Shuwen, D., 1998. U/Pb zircon ages constrain the architecture of the ultrahigh-pressure Qinling-Dabie Orogen, China, *Earth planet. Sci. Lett.,* **161,** 215–230.

Jahn, B.M. *et al.*, 1988. Archaean crustal evolution in China: the Taishan complex and evidence for juvenile crustal addition from long-term depleted mantle, *Precambrian Res.,* **38,** 381–403.

Kirschvink, J.L., 1980. The least-square line and plane and the analysis of palaeomagnetic data, *Geophys. J. R. astr. Soc.,* **62,** 699–718.

Kobayashi, T., 1967. Stratigraphy of the Chosen Group in Korea and South Manchuria and its relation to the Cambro-Ordovician formations of other areas, *J. Fac. Sci., Univ. Tokyo, Sect. 2,* **16,** 381–535.

Li, S. *et al.*, 1993. Collision of the North China and Yangtse Blocks and formation of coesite-bearing eclogites: timing and processes, *Chem. Geol.,* **109,** 89–111.

Lowrie, W., 1990. Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties, *Geophys. Res. Lett.,* **17,** 159–162.

McFadden, P.L., 1990. A new fold test for palaeomagnetic studies, *Geophys. J. Int.,* **103,** 163–169.

Okay, A.I. & Şengör, A.M.C., 1992. Evidence for intracontinental thrust-related exhumation of the ultra-high-pressure rocks in China, *Geology,* **20,** 411–414.

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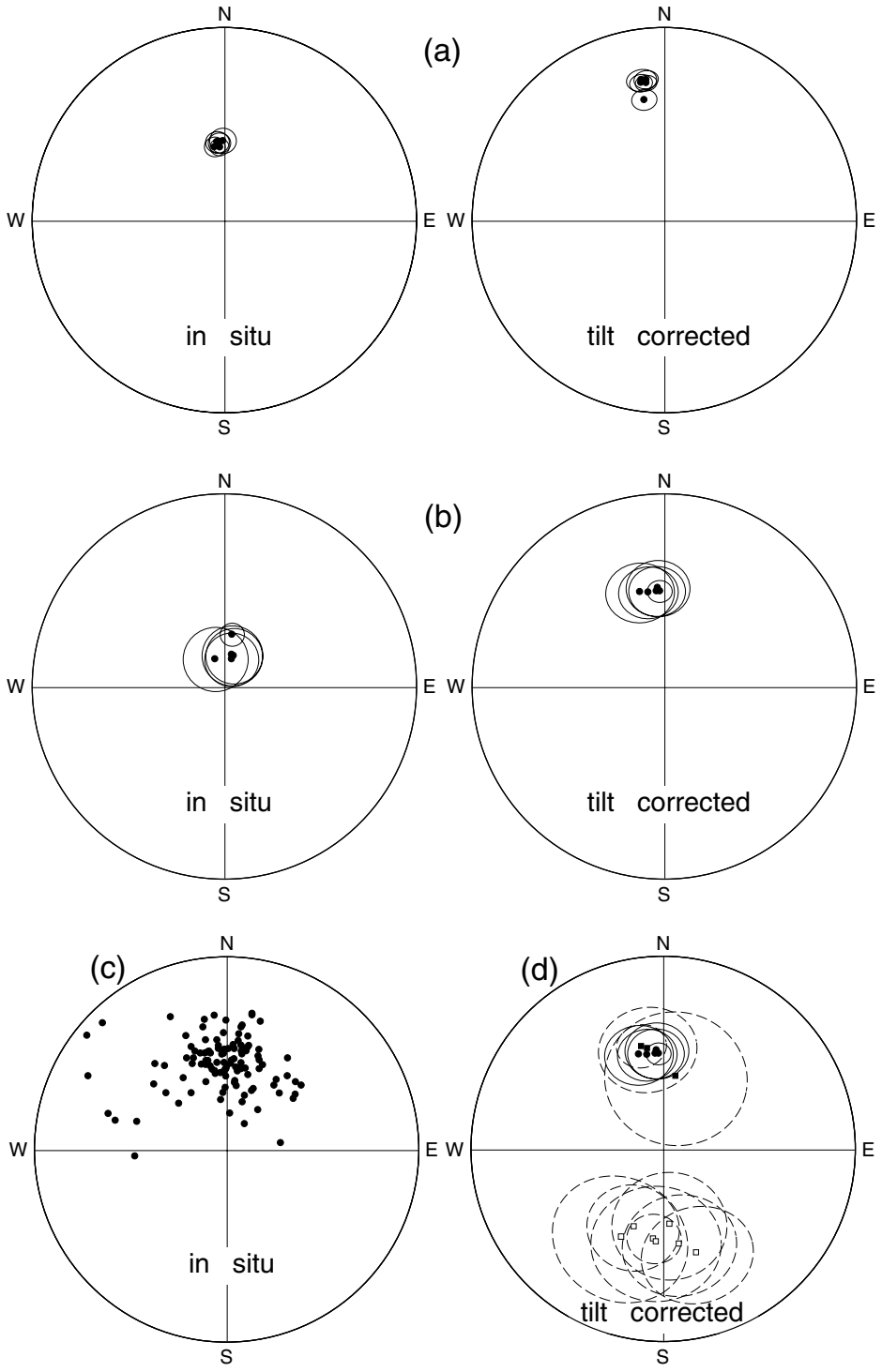


Figure 6. Equal-area projections of directions of the (a) low- and (b) high-temperature components (with associated 95 per cent confidence limits) from five sites of the Early Jurassic Yongdingzhuang Formation in the Datong area (sites DTJ34, 35, 36, 37 and 38). (c) Equal-area projections of directions of the low-temperature component before tilt correction from all sites except nine sites (YAT6, 19, 21, DTJ34, 35, 36, 37, 38 and 39). (d) Equal-area projections of directions of the Early Jurassic Yongdingzhuang Formation in the Datong area (circles, with associated 95 per cent confidence limits) drawn with those of the coeval Fuxian Formation in the Ordos Basin (squares, with associated 95 per cent confidence limits in dashed lines, Yang *et al.* 1992) calculated at the Datong area reference point (40.0°N, 113.0°E).

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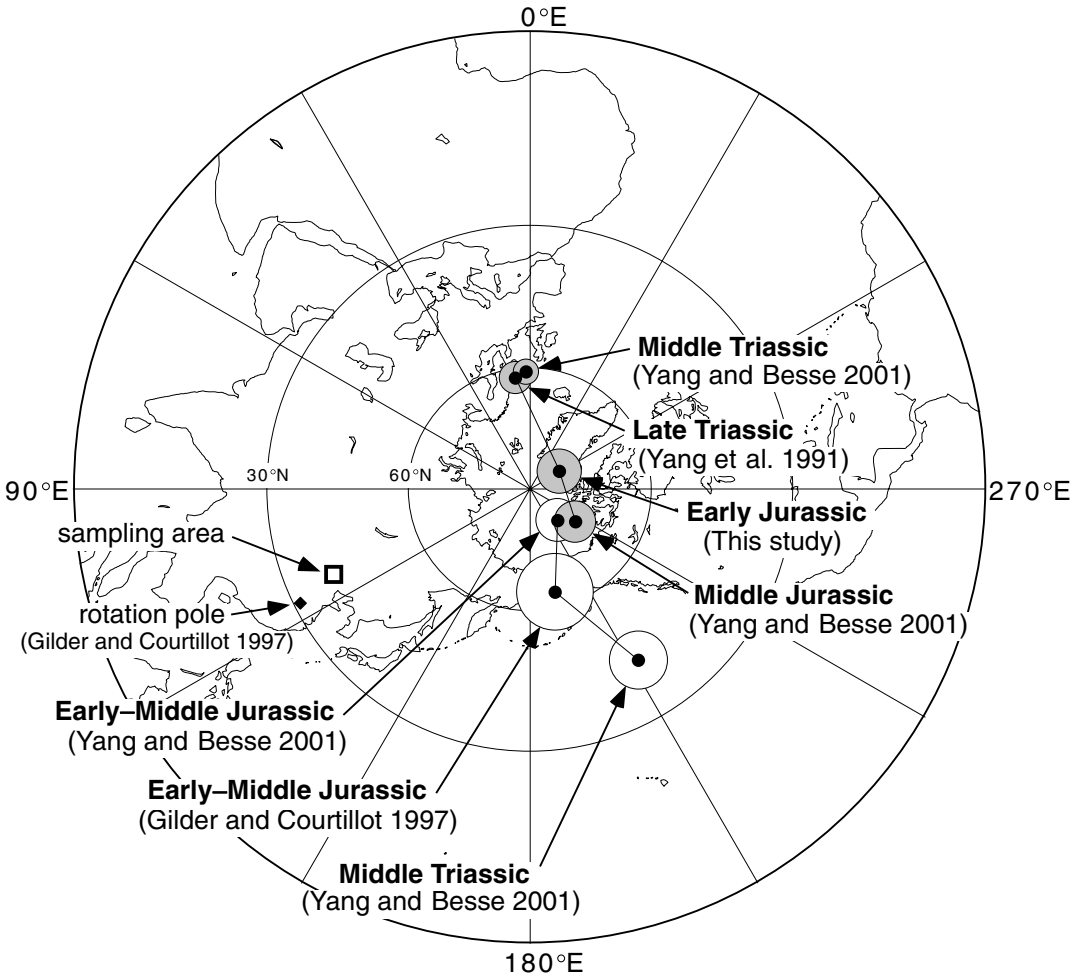


Figure 7. Middle Triassic to Middle Jurassic palaeomagnetic poles (with associated 95 per cent confidence limits) for the North and South China Blocks. The poles for the North (South) China Block show the shaded (open) confidence circles. The open square denotes the reference point of the Datong area. The solid diamond represents the pole of rotation of the North and South China Blocks proposed by Gilder & Courtillot (1997). Refer to Table 3 for data details.

Rowley, D.B., Xue, F., Tucker, R.D., Peng, Z.X., Baker J. & Davis, A., 1997. Ages of ultrahigh pressure metamorphism and protolith orthogneisses from the eastern Dabie Shan: U/Pb zircon geochronology, *Earth planet. Sci. Lett.,* **151,** 191–203.

Song, B., Nutman, A.P., Liu, D.Y. & Wu, J.S., 1996. 3800 to 2500 Ma crustal evolution in the Anshan area of Liaoning Province, northeastern China, *Precambrian Res.,* **78,** 79–94.

Yang, Z. & Besse, J., 2001. New Mesozoic apparent polar wander path for south China: tectonic consequences, *J. geophys. Res.,* **106,** 8493–8520.

Yang, Z., Ma, X., Besse, J., Courtillot, V., Xing, L., Xu, S. & Zhang, J., 1991. Paleomagnetic results from Triassic sections in the Ordos Basin, North China, *Earth planet. Sci. Lett.,* **104,** 258–277.

Yang, Z., Courtillot, V., Besse, J., Ma, X., Xing, L., Xu, S. & Zhang, J., 1992. Jurassic paleomagnetic constraints on the collision of the North and South China Blocks, *Geophys. Res. Lett.,* **19,** 577–580.

Yin, A. & Nie, S., 1993. An indentation model for North and South China collision and the development of the Tan-Lu and Honam fault systems, eastern Asia, *Tectonics,* **12,** 801–813.

Yokoyama, M., Liu, Y., Halim, N. & Otofuji, Y.-i., 2001. Paleomagnetic study of Upper Jurassic rocks from the Sichuan basin: tectonic aspects for the collision between the Yangtze and the North China Block, *Earth planet. Sci. Lett.,* **193,** 273–285.

Zhao, X. & Coe, R., 1987. Palaeomagnetic constraints on the collision and rotation of the North and South China, *Nature,* **327,** 141–144.

Zhao, X., Coe, R.S., Gilder, S.A. & Frost, G.M., 1996. Palaeomagnetic constraints on the palaeogeography of China: implications for Gondwanaland, *Australian J. Earth Sci.,* **43,** 643–672.

Zheng, Z., Kono, M., Tsunakawa, H., Kimura, G., Wei, Q., Zhu, X. & Hao, T., 1991. The apparent polar wander path for the North China Block since the Jurassic, *Geophys. J. Int.,* **104,** 29–40.

Zijderveld, J.D.A., 1967. AC demagnetization of rocks: analysis and results, in *Methods in Paleomagnetism,* pp. 254–286, edsCollinson, D.W., Creer, K.M. & Runcorn, S.K., Elsevier, Amsterdam.

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